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BIOMEDICAL LIMITED

INSTALLATION QUALIFICATION (IQ)

PROTOCOL
FOR
NON-PVC FILM AUTOMATIC BAG
FORM-FILL-SEAL MACHINE

EQUIPMENT SR. NO. –

EQUIPMENT ID. –

LOCATION – INFUSION FILLING ROOM

PROTOCOL No. – BML/PDF/FR/PT/25/001

PROVIDED BY:

BIOMEDICAL LIMTED

KWARA STATE, NIGERIA

CONTROL COPY

TABLE OF CONTENT

Cover Page	1
Table of content	2
Protocol Format Approval Sheet	3
Objective	4
Scope	5
Responsibilities	5
Equipment identification and description	5
Prerequisite	6
Abbreviations and definitions	6
References	7
Equipment installation Checklist	7-17
Installation Qualification Report	17
Deviation Report	18
Conclusion	19
List of attachments	19
Report Approval	20

1.0 PROTOCOL FORMAT APPROVAL:

Signatures in this page indicate that this document and its format have been prepared, checked and approved by the personnel concerned. It shall be signed and date as shown below

	WRITTEN:	**REVIEWED:**
Signature:		
Position:		
Date:		

	APPROVED	**AUTHORIZED**
Signature:		
Position:		
Date:		

2.0 OBJECTIVE

The purpose of the study is to ensure proper installation of the two **NON-PVC FILM AUTOMATIC BAG FORM-FILL-SEAL MACHINE** at **THE FILLING ROOM, INFUSION FACTORY, BIOMEDICAL LIMITED**

<u>This study verifies:</u>

- Qualification of the equipment, re installation and to ensure that all background information is performed to assure conformance to **BIOMEDICAL LIMITED** and manufacturer's specifications.
- The equipment meets the current good manufacturing practices requirements and all other regulatory obligations
- No unauthorized or unrecorded modifications have taken place.
- Operation procedures/preventive maintenance procedures have been identified and listed.
- All critical instrumentation has been identified
- All utilities are properly connected
- All safety features are accounted for
- All relevant data gave have been checked

3.0 SCOPE

- This protocol is applicable in carrying out the installation qualification of the two Form Fill Seal Machine at the Infusion manufacturing site, Biomedical Limited, Kwara State.
- To be performed after the completion of the design qualification
- To be performed at the time of relocation.

4.0 RESPONSIBILITY AND QUALIFICATION TEAM

The following departments shall be involved in:

DEPARTMENT	RESPONSIBILITY
Engineering Team	Provision of needed utility and machine installation
Head Of Engineering	Review of the Installation Qualification Document
QA –Manager	Final Documentation and system acceptability
Superintendent Pharmacist/factory Manager	Authorization

The following personnel will be involved in the execution of the Qualification

S/N	DEPARTMENT	NAME	SIGNATURE
1			
2			
3			
4			
5			

5.0 EQUIPMENT IDENTIFICATION AND DESCRIPTION

IDENTIFICATION

Equipment Name	NON-PVC FILM AUTOMATIC BAG FORM-FILL-SEAL MACHINE
Manufacturer's Name	SHANGAI IVEN PHARMATECH ENGINEERING CO., LTD, CHINA
Supplier's name	SHANGAI IVEN PHARMATECH ENGINEERING CO., LTD, CHINA
Location of Installation	FILLING ROOM (INFUSION FACTORY)

DESCRIPTION:

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6.0 PREREQUISITE

The equipment shall confirm to the specifications and requirements as specified in the Design Qualification

S/N	ACCEPTANCE CRITERIA
1	User Requirement specification
2	Design Qualification

7.0 ABBREVIATIONS AND DEFINITIONS

Abbreviation	Definition
SOP	Standard Operating Procedure
MOC	Material of Construction
CO	Company
ENGR	Engineering
IQ	Installation Qualification
QA	Quality Assurance
VAC	Voltage Alternating Current
S. S	Stainless Steel
LTD	Limited
°C	Degree Celsius
V	Voltage
GMP	Good manufacturing practices
AC	Alternating Current
Amps	Ampere
W	Watt

Mm	Millimeter
VDC	Voltage Direct Current
DC	Direct Current
PVC	Polyvinyl Chloride
MCB	Miniature Circuit Breaker
MCCB	Molded case circuit breaker
KW	Kilowatt
PLC	Programmable Logic Control

8.0 REFERENCES:

The list of reference documents, which are referred for preparation and review of this document, is as follows: -

- S.O.P for Installation, Operation & Performance Qualification of equipment & Instrument: BML-QDG-QA-111
- Machine Manual supplied by SHANGAI IVEN PHARMATECH ENGINEERING CO., LTD
- Design Qualification supplied by SHANGAI IVEN PHARMATECH ENGINEERING CO., LTD

9.0 EQUIPMENT INSTALLATION CHECKLIST:

Pre-Installation Consideration

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	There is no observable physical damage to the machine and all the contact parts are free from scratches & damages.			
2	Sufficient room for servicing, handling & maintenance is provided (around the machine)			
3	Check the floor space. It should not be inclined.			

Assembly of non-assembled items

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	Assemble/connect the machine Laminar air flow hood to the main machine assembly			
2	Assemble/connect the HEPA Filters to the main machine assembly			
3	Assemble/connect the machine Laminar air flow hood prefilters to the main machine assembly			
4	Assemble/connect the machine Perspex doors to the main machine assembly			

Utilities

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	Power Requirement (Three-phase, 5 wires, L1, L2, L3, Neutral & Earth 415VAC ±10% 50Hz)			
2	Adequate illumination at the Proposed Machine room			
3	Proper Earthing System available			

Equipment compliance

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	Machine name plate present			
2	Machine Model Number: SRD4-3600BPH			
3	Machine Factory Number: 20244755-202R			
4	Power Rating: 32KW			

Safety compliance

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	All external surfaces are free of sharp edges.			
2	Safety earthling arrangement is available.			
3	Emergency Stop Buttons available			
4	Electrical disconnect available			

Electrical diagram check

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	Verify the electrical diagram			
2	Check whether the position of each component in the electrical is correct			
3	Check whether the wiring of the equipment is consistent with electrical diagrams			
4	Check if the electrical diagram is attached to the operation manual			

P&ID check

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	Check that the filling pipe system is consistent with the P&ID drawing			
2	Check that the supplier and model number of each component match			
3	Check that the component installed are consistent with the P&ID part list			
4	Check that the P&ID is attached to the operation manual			

List of Main Components with their material of construction

S/N	Component	Origin/ Configuration	Qualified / Not Qualified	Checked By	Verified By
MECHANICAL PART LIST					
1	Compact cylinder	Location: Film feeding station, Brand: Festo, Specification: ADN-20-25-I-P-A			
2	Brake Motor	Location: Film feeding station, Brand: Sew, Specification: KAF37DRN71MS4/BE03/TH			
3	Ionic wind bar	Location: Film feeding station, Brand: SMC, Specification: ZS41-340P-06B			
4	Compact double force cylinder	Location: Printing station, Brand: Festo, Specification: ADNH-100-50-A-P-A-2N-100K8			
5	Standard cylinder	Location: Printing station, Brand: Festo, Specification: ADN-16-10-I-P-A			
6	Brake motor	Location: Printing station, Brand: Sew, Specification: WAF30 DR63L4/BR/THWAF30DRN71MS4/BE03/TH			
7	Parallel gripper	Location: Film Pulling, Brand: Festo, Specification: DHPS-25-A			
8	Double piston-cylinder	Location: Film Pulling, Brand: Festo, Specification: DPZJ-32-50-P-A-S2			
9	Electric cylinder	Location: Film Pulling, Brand: Festo, Specification: ELGA-TB-KF-80-830-0H			
10	Parallel gripper	Location: pre heating, Brand: Festo, Specification: DHPS-35-A			
11	X Eliminator	Location: Bag Transferring, Brand: SMC, Specification: IZN10E-11P06Z-B1			
12	Cylinder with guide rod	Location: Bag making station, Brand: Festo, Specification: DFM-25-30-P-A-GF			
13	Double piston cylinder	Location: Bag making station, Brand: Festo, Specification: DPZJ-25-25-P-A-S2			

14	Cylinder with guide rod	Location: Bag making station, Brand: Festo, Specification: DFM-32-20-P-A-GF			
15	TOX buster Cylinder	Location: Bag making station, Brand: TOX, Specification: EL8.160.50.6D			
16	Pneumatic linear drive unit	Location: port feeding, Brand: Festo, Specification: DGPL-32-250-PPV-A-B-KF-GK-SH-GA			
17	Hydraulic buffer	Location: port feeding, Brand: Festo, Specification: YSR-12-12-C			
18	Medium drive guider	Location: port feeding, Brand: Festo, Specification: DFM-50-50-P-A-GF			
19	Straight ball guide pneumatic drive unit	Location: port feeding, Brand: Festo, Specification: DGPL-25-185-PPV-A-KF-B			
20	Gripper	Location: port feeding, Brand: Festo, Specification: DHPS-25-A			
21	Cylinder	Location: port feeding, Brand: Festo, Specification: ADNGF-25-15-P-A			
22	X eliminator	Location: port feeding, Brand: SMC, Specification: IZN10E-11P06Z-B1			
23	Cylinder	Location: port welding, Brand: Festo, Specification: DFM-63-25-P-A-GF			
24	Rotating gripper	Location: port welding, Brand: Festo, Specification: DHRS-25-A			
25	Circular cylinder	Location: port welding, Brand: Festo, Specification: DSNU-20-50-PPV-A			
26	Hydraulic buffer	Location: transferring station, Brand: Festo, Specification: DYSC-7-5-Y1F			
27	Cylinder with guide rod	Location: transferring station, Brand: Festo, Specification: DFM-32-20-P-A-GF			
28	Double rod cylinder	Location: transferring station, Brand: Festo, Specification: DPZJ-20-80-P-A-S2			
29	Double rod cylinder	Location: transferring station, Brand: Festo, Specification: DPZJ-25-80-P-A-S2			
30	Rodless cylinder	Location: transferring station, Brand: Festo, Specification: DGPL-40-450-PPV-A-B-KF-GK-SH-GA			

31	Rotating Cylinder	Location: transferring station, Brand: Festo, Specification: DSM-25-270-P-A-B			
32	Hydraulic Buffer	Location: transferring station, Brand: Festo, Specification: DG-GA-40-YSR			
33	Compact Guide Cylinder	Location: filling, Brand: Festo, Specification: ADNGF-100-30-P-A			
34	Double Rod Cylinder	Location: filling, Brand: Festo, Specification: DPZJ-32-40-P-A-S2			
35	Double Acting Cylinder	Location: filling, Brand: Festo, Specification: DPZJ-32-50-P-A-S2			
36	Hydraulic Buffer	Location: filling, Brand: Festo, Specification: YSR-12-12-C			
37	Steam Trap	Location: filling, Brand: customized, Specification: 1811-DN20,1/8"			
38	Compact cylinder	Location: cap sealing, Brand: Festo, Specification: ADNGF-32-30-P-A			
39	Toothed belt electric cylinder	Location: cap sealing, Brand: Festo, Specification: ELGA-TB-KF-80-300-0H			
40	Double piston cylinder	Location: cap sealing, Brand: Festo, Specification: DPZ-32-40-P-A			
41	Standard cylinder	Location: cap sealing, Brand: Festo, Specification: DSBC-32-100-PPVA-N3			
42	Double piston cylinder	Location: cap sealing, Brand: Festo, Specification: DPZJ-20-25-P-A-S2			
43	X Eliminator	Location: cap sealing, Brand: SMC, Specification: IZN10E-11P06Z-B1			
44	Rotating gripper	Location: Output station, Brand: Festo, Specification: DHWS-32-A			
45	Rodless Cylinder	Location: Output station, Brand: Festo, Specification: DGPL-40-750-PPV-A-B-KF-GK-SV-GA			
46	Hydraulic Buffer	Location: Output station, Brand: Festo, Specification: YSR-16-20-C			
47	European standard cylinder	Location: Output station, Brand: Customized, Specification: DNC06013			

CRITICAL SENSORS					
48	Mass flow meter	Function: Measure filling volume, Vendor: Endress + Hauser, Model: 8BE08-AFTSABAB41A3, Range: 0.1g-∞, Accuracy: 0.1			
49	Thermocouples	Function: Measure die Temperature, Vendor: Tianjin Zhonghuan, Model: RSY-DO/K-01, Range: -200-1250°C, Accuracy: 1.1°C			
50	Thermocouples	Function: Measure die Temperature, Vendor: Tianjin Zhonghuan, Model: RSY-DO/K-04, Range: -200-1250°C, Accuracy: 1.1°C			
51	Thermocouples	Function: Measure die Temperature, Vendor: Tianjin Zhonghuan, Model: RSY-DO/K-02, Range: -200-1250°C, Accuracy: 1.1°C			
52	Magnetic switch	Function: Detect Cylinder position, Vendor: IFM, Model: MK5101, Range: NA, Accuracy: NA			
53	Inductive Transducer	Function: Detect position, Vendor: IFM, Model: IE5329, Range: NA, Accuracy: NA			
54	Photoelectric Switch	Function: Detect position, Vendor: BANNER, Model: Q12AB6FF30, Range: 0-30mm, Accuracy: NA			
55	Photoelectric Switch	Function: Detect material level in the hopper, Vendor: BANNER, Model: QS30AF, Range: 0-30cm, Accuracy: NA			
56	Photoelectric sensor	Function: Detect bag, Vendor: BANNER, Model: QS18VP6XLPQ7, Range: NA, Accuracy: NA			
57	Mutual Inductor	Function: Measure heating plate current, Vendor: Tianjin Zhonghuan, Model: TCYT, Range: 0-200A, Accuracy: 1A			

58	Temperature sensor	Function: Measure SIP Temperature, Vendor: Zibo Feiyan, Model: XH-PT2-143122-150, Range: -50-200°C, Accuracy: 1.1°C			
59	Flow switch	Function: Detect cooling water, Vendor: SENTINEL, Model: PLSX-1/2G-AP6D-SC, Range: 0-5 m/s, Accuracy: NA			
60	Pressure sensor	Function: Detecting negative pressure gas, Vendor: SMC, Model: ZSE40AF-01-Y-MA, Range: -1-16 bar, Accuracy: 1.5%FS			
61	Pressure sensor	Function: Detect gas source Pressure, Vendor: Festo, Model: SPAU-P10R-W-G18FD-L PNLK-PNVBA-M8U, Range: 0-10 bar, Accuracy: 1.5%FS			
ELECTRICAL HARDWARE LIST					
62	PLC	Function: Handle all program running events as the main controller, Vendor: SIEMENS, Order Number: 6ES7512-1CK01-0AB0			
63	Digital Input Module (DI)	Function: Read digital input signal, Vendor: SIEMENS, Order Number: 6ES7521-1BL00-0AB0			
64	Digital Output Module (DQ)	Function: Output digital input signal, Vendor: SIEMENS, Order Number: 6ES7532-5HD00-0AB0			
65	Analog Input Module (AI)	Function: Read analog input signal, Vendor: SIEMENS, Order Number: 6ES7531-7QF00-0AB0			
66	Analog Output Module (AQ)	Function: Output analog input signal, Vendor: SIEMENS, Order Number: 6ES7532-5HD00-0AB0			
67	Memory Card (SD)	Function: Store Program, Vendor: SIEMENS, Order Number: 6ES7954-8LE03-0AA0			

68	Touch screen (HMI)	Function: Human Machine Interface, interactive data with controller, Vendor: SIEMENS, Order Number: 6AV2124-0MC01-0AX0			
69	Switchboard	Function: Exchange data for various controller, Vendor: SIEMENS, Order Number: 6GK5008-0BA10-1AB2			
70	Switching Power Supply (PW)	Function: Provide power supply, Vendor: MEANWELL, Order Number: SDR-960-24			
71	Relay (KA)	Function: Isolate power supply and extend contacts, Vendor: OMRON, Order Number: MY2NJ-24V			
72	Solid State Relay (GK)	Function: Control output in temperature control, Vendor: CYDON, Order Number: KSI240D25-LM(068)			
73	Safety Relay (KNA)	Function: Provide personnel and equipment in case of emergency, Vendor: Schmersal, Order Number: SRB301MC-24V			
74	Silicon Controlled Rectifier Power Regulator (KKG)	Function: Provide the required current value to the heater chip Vendor: CYDON, Order Number: CRB480H75P;75A;480VAC			
75	Molded case circuit breaker (QF)	Function: Connect and disconnect circuit, Vendor: SCHNEIDER, Order Number: C11E6TM063BZ			
76	Contactor (KM)	Function: isolate power supply and extends contact, Vendor: SCHNEIDER, Order Number: LC1D12BDC LC1D38BDC			
77	Miniature Circuit breaker (QF)	Function: connect and disconnect circuit, Vendor: SCHNEIDER, Order Number: OSMC32N3C16 OSMC32N3C40 OSMC32N1C10 OSMC32N1C16 OSMC32N1C1 OSMC32N1C4 OSMC32N2D16 OSMC32N2C25			

78	Motor Starter (Q)	Function: protect the motor against phase loss, Vendor: SCHNEIDER, Order Number: GV2ME10C GV2ME07C			
79	Frequency Converter (BP)	Function: Adjust the motor frequency speed, Vendor: SCHNEIDER, Order Number: ATV310H037N4A			
80	Controller (SF)	Function: Precise control of motor positioning, Vendor: LENZE, Order Number: E70ACMSE0104SA2ETE			
81	Valve Terminal (FD)	Function: Provide Input Signal and control cylinder action, Vendor: Festo, Order Number: 45P-R-W-LVMX-8EU8EU-32M			
LAMINAR AIR FLOW HOOD COMPONENTS					
82	Air Blower	Model: EBM355/310 Fan, MOC: SS 304, Manufacturer: EBM, Quantity: 28 Pieces			
83	LED Light	Model: T5, Manufacturer: Philips, Quantity: 56 Pieces			
84	Differential Pressure Gauge	Model: 0-500A/DM-105-SS, Manufacturer: Dwyer Quantity: 12 Pieces			
85	High Efficiency Filter (H14)	Dimensions: 961*569*90mm (2 Pieces) 961*506*90mm (5 Pieces) 961*656*90mm (1 Piece) 961*545*90mm (1 Piece) 743*468*90mm (2 Pieces) 961*553*90mm (2 Pieces) 791*468*90mm (2 Pieces) 645*403*90mm (4 Pieces) 941*403*90mm (2 Pieces) 821*612*90mm (4 Pieces) 821*447*90mm (2 Pieces) 821*417*90mm (2 Pieces) Manufacturer: IVEN			
86	PAO Injection Port	MOC: SS 304, Manufacturer: IVEN, Quantity: 28 Pieces			
87	PAO Test Port	MOC: SS 304, Manufacturer: IVEN, Quantity: 28 Pieces			

88	Aluminum frame, black nylon mesh	525*345*10 (4 Pieces) 740*345*10 (2 Pieces) 1090*345*10 (1 Pieces) 940*345*10 (1 Pieces) 640*345*10 (7 Pieces) 1040*345*10 (1 Pieces) 540*345*10 (2 Pieces) 840*345*10 (1 Pieces) 500*345*10 (2 Pieces) 920*345*10 (4 Pieces)			

10.0 INSTALLATION QUALIFICATION REPORT:

Attachment of all reports written as:

#Attachment 1

#Attachement 2 etc.

Date of Installation:

Date of Completion:

11.0 DEVIATION REPORT

Prepare a Deviation report including the justification of acceptance and impact on the function of the system/equipment

DEVIATIONS:
JUSTIFICATION FOR ACCEPTANCE:
IMPACT ON OPERATION:

Written By: **Date:**

Reviewed By: **Date:**

Approved By: **Date:**

12.0 CONCLUSION

13.0 List of Attachment

Attach all completed, signed data record forms

14.0 REPORT APPROVAL

Written By: **Date:**

Reviewed By: **Date:**

Approved By: **Date:**